

Trilaterales Transferprojekt

DFG – Fraunhofer

Ausschreibung 2022/23

Projekttitel

Allocation and Pricing in Electricity Markets

Akronym (max. 15 Zeichen)

APEM

Name der Projektleiterin/des Projektleiters und zugehörige Institution (Universität/HAW/FH)

Prof. Dr. Martin Bichler, Technische Universität München (TUM)

Name der Projektleiterin/des Projektleiters und zugehörige Institution mit Geschäftsfeld (Fraunhofer)

Prof. Dr. Martin Weibelzahl, Fraunhofer Institute for Applied Information Technology FIT (FIT)

Anwendungspartner (Unternehmensbezeichnung inkl. vollständiger Adress- und Kontaktdaten, Website)

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Beteiligungen am Anwendungspartner

Keine Beteiligung

Projekttitel des DFG-Vorgängerprojekts

Preissetzung auf kombinatorischen Märkten

DFG-Geschäftszeichen und Laufzeit

BI 1057/8-1, 01.04.2018 – 31.12.2022

Projektleiterin/Projektleiter DFG-Vorgängerprojekt

Prof. Dr. Martin Bichler

Wissenschaftliche Mitarbeiterinnen/Mitarbeiter DFG-Vorgängerprojekt

Dr. Stefan Waldherr

Beschreibung des Vorhabens – Projektanträge / Description of the project - project applications

Bichler, Martin, TU München
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Allocation and Pricing in Electricity Markets

Beschreibung des Vorhabens / Description of the project

1 Ausgangslage / Status Quo

Stand der Forschung und eigene Vorarbeiten / State of research and preliminary work

This project studies allocation and pricing in complex electricity markets, focusing on the two highly relevant aspects of pricing with non-convexities and optimal power flow. While pricing on markets with multiple objects has been central to the economic sciences for a long time, the possibility to solve large non-convex allocation problems in fields such as industrial procurement or in spectrum sales has led to new questions about pricing. Electricity markets are probably the most important type of non-convex markets, and they exhibit additional complexities due to the coupled nature of different markets, i.e., physical transmission networks that need to be considered for clearing and pricing.

The study of *competitive equilibria* is well-established in economic theory and led to the celebrated welfare theorems (Arrow and Debreu 1954). These theorems state that in markets with convex preferences, a competitive equilibrium will maximize social welfare and that every welfare-maximizing allocation can be supported by competitive equilibrium prices. At those prices no market participant has incentives to deviate from the welfare-maximizing allocation, and the outcome is envy-free and budget-balanced. More recently, competitive equilibrium theory has focused on markets with indivisible goods and conditions for the existence of *Walrasian equilibrium prices*, i.e., linear and anonymous competitive equilibrium prices (Bikhchandani and Mamer 1997, Gul and Stacchetti 1999, Bikhchandani and Ostro 2002, Baldwin and Klemperer 2019).

Unfortunately, conditions for the existence of Walrasian equilibria are very restrictive and most real-world markets have non-convex value and cost functions, leading to non-convex welfare maximization problems. Electricity markets are a prime example of such non-convex markets as described before, but similar characteristics can be found in spectrum auctions or markets for environmental access rights (Bichler, Goeree et al. 2014, Bichler, Fux et al. 2018).

One possibility to deal with the non-existence of Walrasian equilibria is to deviate to non-linear and personalized competitive equilibrium prices. However, there might exist exponentially many prices, and personalized prices convey little information about the scarcity of goods. Non-convex markets in practice typically require linearity and anonymity of prices. For example, in electricity markets the spot market prices serve as a baseline for, e.g., over-the-counter trades and futures markets. Non-linear and personalized prices are inadequate for these purposes. Another possibility to deal with non-existence of Walrasian equilibria is to deviate from the welfare-maximizing outcome and find near-equilibria (Starr 1969, Milgrom and Watt 2022). While current European day-ahead electricity markets accept welfare losses for linear prices, wholesale electricity markets in the U.S. implement optimal outcomes and determine linear and anon-

ymous market prices that are complemented by personalized out-of-market side-payments. These payments compensate for possible losses of market participants, or they mitigate incentives to deviate from the efficient outcome by paying lost opportunity costs. Similar pricing rules are now also being investigated for European day-ahead electricity markets¹. However, the current pricing rules in U.S. electricity markets, such as Integer Programming Pricing (O'Neill, Sotkiewicz et al. 2005) or Convex Hull Pricing (Hogan and Ring 2003, Gribik, Hogan et al. 2007), have come under scrutiny since the side-payments to some of the market participants are often high². While this is an important practical problem for market operators in Europe and the U.S., it represents at the same time a fundamental problem in the economic sciences beyond electricity markets: *How can goods be priced in markets with non-convex preferences?*

Apart from non-convexities, the computed allocation on electricity markets must respect transmission constraints of the electricity grid. With an accurate representation of the transmission network and the underlying physics, this can be described as a non-linear and non-convex optimization problem referred to as Alternating Current Optimal Power Flow (ACOPF) problem (Molzahn and Hiskens 2019). This ACOPF is computationally intractable for the problem sizes that we observe in real-world electricity markets. As a result, linearized network models such as the Direct Current Optimal Power Flow (DCOPF) problem are used in practice as an approximation. However, the optimal solution for DCOPF is generally neither ACOPF-optimal nor ACOPF-feasible. This requires market operators or transmission system operators (TSOs) to adjust the dispatch after the market clearing to reach a physically feasible outcome. The failure to account for physical grid constraints leads to high redispatch costs.

In recent years, substantial research has been devoted to finding tighter convex relaxations and approximations for the ACOPF problem (Molzahn and Hiskens 2019). This research is driven by advances in convex non-linear optimization algorithms and focuses on the optimality of the solution and computational costs to obtain it. However, there is little insight on the impact on prices. The U.S. Federal Energy Regulation Commission (FERC) suggests that the size and concentration of side-payments can be affected by the inability to accurately model ACOPF, and that the benefit-to-cost ratio of more accurate power flow representations is at least 100-fold (Cain and O'Neill 2012).

We want to derive prices that best reflect marginal costs on a network node in a coupled electricity market with non-convex preferences of participants. This question has regained interest due to the strongly increasing levels of renewable energy sources, which have different cost characteristics. We want to explore the impact of different approximations and convex relaxations of the underlying ACOPF on prices and compare such nodal prices to the zonal prices currently in use in Europe. While the fundamental question of pricing in non-convex markets has received significant attention in the literature, the specifics of coupled electricity markets (e.g., transmission constraints between nodes) provide new challenges for fundamental research that have often been ignored. We want to understand the scope for which the welfare theorems hold in such markets and to which extent envy-freeness and budget balance can be satisfied in coupled and non-convex markets.

¹ https://www.nemo-committee.eu/assets/files/NEMO_CACM_Annual_Report_2020_deliverable_1_pub.pdf

² <https://www.ferc.gov/industries-data/electric/electric-power-markets/energy-price-formation>

Universität / University

Our DFG project BI 1057/8 "Pricing in Combinatorial Exchanges" (2018-2022) addressed fundamental questions of pricing in combinatorial markets. Competitive equilibrium theory largely assumes that the underlying allocation problem is a convex welfare maximization problem. If convexity is violated, we are lacking a principled way to compute prices. As outlined earlier, most real-world markets have non-convex value and cost functions leading to non-convex and computationally hard welfare maximization problems. Electricity markets are a prime example of such non-convex markets.

The DFG project BI 1057/8 focused on the question how prices can be computed in combinatorial exchanges. Ideally, at those prices, no participant would want to deviate from the welfare-maximizing allocation, no one makes a loss, and the market is budget-balanced, requiring no side-payments. It is impossible to satisfy all of these properties with anonymous and linear Walrasian prices if the welfare maximization problem is non-convex in general.

Our work schedule was structured along 6 work packages (WPs). WP 1a and 1b focused on linear prices in non-convex markets, the underlying theory and computational approaches to approximate competitive equilibrium prices. This line of research is most relevant to our transfer project and it resulted in a number of articles, which provide a foundation for our planned work program.

A first article (Bichler, Fux et al. 2018) laid the foundation and describes the linear pricing rules in the presence of non-convex preferences. A second article (Bichler, Fichtl et al. 2020) analyzes characteristics of valuation functions, which allow for Walrasian equilibria. In particular, we leverage convex conjugates and duality theory to explain pricing rules in convex markets as well as auction algorithms from an optimization perspective. A third article (Baldwin, Bichler et al. 2022) focuses on the strong substitutes condition, a sufficient condition for the existence of Walrasian prices. Unfortunately, this condition is hard to grasp and difficult to implement in a bid language. We contributed a new characterization via Minkowski differences that allowed us to define a bid language and efficient algorithms to compute prices. Other articles (Goetzendorff, Bichler et al. 2018, Bichler, Grimm et al. 2020, Bichler, Knörr et al. 2022) deal with linear prices in different application domains. There is no one-size-fits-all approach and the pricing needs to be adapted to the bid language and the allocation problem. The results of project BI 1057/8 provide respective analyses for environmental markets, spectrum sales, and renewable energy auctions. These articles provide foundations for the design of linear pricing rules in electricity spot markets as well, where we also need to consider transmission constraints.

WP 2a and 2b of DFG project BI 1057/8 analyzed non-linear prices, the theory and computational methods. These questions are somewhat related but not at the core of the transfer project. Resulting papers discuss equilibrium strategies in first-price split-award auctions, while one article deals with non-linear pricing in network procurement. Finally, WP 3a and 3b deal with budget constraints in markets (Bichler and Waldherr 2021). Such constraints make the allocation and pricing problem computationally intractable (Σ_2^P -hard) as we show. We published two papers on advertising markets and analyzed bidding strategies and truthful auction design in such markets. In our transfer project on electricity markets, we assume standard quasi-linear preferences where budget constraints do not matter.

Apart from these and other theoretical treatments of pricing in non-convex markets outlined above, we have also deployed specific pricing rules for trading fishery access rights in the field

(Bichler, Fux et al. 2018). The project implemented a combinatorial exchange for more than 600 fishers, and is considered a successful application of a large-scale combinatorial exchange in the field. The work received the EURO Excellence in Practice Award and was also a runner-up for the INFORMS Wagner Prize.

The DFG project BI 1057/8 did not focus on a particular domain, but on pricing in combinatorial exchanges in general. Electricity markets provide a highly relevant, but also complex application domain, where we bring the theoretical insights from BI 1057/8 to bear. Electricity spot markets are different from other non-convex markets due to their specific types of non-convexities, the need for balanced markets at all times, and the need to consider transmission constraints for allocation and pricing. If there is congestion on a link, prices on adjacent nodes shall be different (Grimm, Martin et al. 2016, Weibelzahl and März 2018). There is a long history of research on pricing in electricity markets (Hogan and Ring 2003, O'Neill, Sotkiewicz et al. 2005, Gribik, Hogan et al. 2007, Liberopoulos and Andrianesis 2016, Schiro, Zheng et al. 2016, Knueven, Ostrowski et al. 2022). Most of these pricing heuristics – as they are used in practice – aim to minimize lost opportunity costs. However, these rules are now under scrutiny by the US Federal Energy Regulatory Commission (FERC)³. In particular, the side-payments paid to some market participants are unreasonably high. Overall, this leads to biased market signals. The same questions are at the core of the current revision of the EU market rules⁴, which explores non-uniform pricing for the European day-ahead electricity market.

In our recent research on electricity markets (Ahunbay, Bichler et al. 2022, Bichler, Knörr et al. 2022), we have addressed some of these questions and suggested new pricing rules. These rules were tested on a small and stylized benchmark dataset, the IEEE RTS System with 24 nodes. While our initial experiments are useful as an illustration and confirm the theoretical properties (TRL 4), the challenges of electricity market pricing come with the scale of the problems and the technical details involved. The European electricity grid has *more than 25.000 nodes*. In addition, the small-scale experiments do not consider technical constraints of the generators and the network (e.g., voltage constraints, thermal line limits, technical generation constraints, etc.) or European-like bidding formats. These details can make a substantial difference and results from stylized and small examples do not necessarily carry over. For external validity of the analyses and reliable policy advice, such technicalities need to be considered. Ultimately, policy makers need to understand the trade-offs in electricity market design with respect to welfare losses or uplifts that need to be paid to participants.

Fraunhofer

FIT has extensive expertise in electricity markets. For example, FIT leads one of the largest EU-funded research projects, [OneNet](#), on the future of the European electricity system. As a steering-committee member, Prof. Weibelzahl plays a leading role in the [Kopernikus-Project SynErgie](#) (funded by the Federal Ministry of Education and Research since 2016 with up to 100 million euros). Within SynErgie and an (inter-)national network of market design experts, we regularly discuss general market-design questions on a higher level with the aim of enabling active consumer participation on electricity markets. Key outcomes were a series of [whitepapers](#) discussing more general topics around nodal pricing, secondary markets, and investment signals. Prof. Bichler from TUM is one of the PIs in this project. In this context, several research papers were published that deal with pressing electricity-market design questions (Weibelzahl and März

³ <https://www.ferc.gov/industries-data/electric/electric-power-markets/energy-price-formation>

⁴ https://www.nemo-committee.eu/assets/files/NEMO_CACM_Annual_Report_2020_deliverable_1_pub.pdf

2018, Fridgen, Michaelis et al. 2020, Bichler, Buhl et al. 2022, Grimm, Martin et al. 2022, Heffron, Körner et al. 2022, Bjørndal, Bjørndal et al. 2023).

With its in-depth electricity-market know-how, FIT and particularly its branch Business & Information Systems Engineering (FIT BISE) in Augsburg and Bayreuth are also involved in various consultancy projects (e.g., with different TSOs), regularly advise European (and German) regulators, and are active in the [European bidding zone review](#) (BZR) as a member of the so-called Bidding Zone Review Consultative Group. The latter is the central expert platform for ENTSO-E and its members to achieve the objectives of the BZR. Only last summer, FIT BISE and TUM jointly organized a [summit](#) with responsible representatives of the European Commission, ACER, and ENTSO-E to work on pressing market-design challenges.

Besides the European level, FIT BISE is for instance involved in the development of the current German redispatch within the project Redispatch 3.0. FIT BISE also actively works on the market integration of decentral assets in Germany in the project VIdES, which is funded by the German Energy Agency (DENA). In such projects, FIT regularly implements different optimization models and develops KPIs to analyze and evaluate market design options, closely collaborating with relevant industry partners and other stakeholders. As further described below, with the transfer project, FIT wants to strengthen its competences and be able to perform realistic simulations that will inform policy makers and market designers about the right choices for a future-proof electricity market design.

Anwendungspartner / Application partners

Legal Form	TenneT TSO GmbH
Year of foundation	2009
Company headquarters	Bayreuth
Head of the company	Maarten Abbenhuis, Tim Meyerjürgens, Dr. Arina Freitag
Employees	~ 3.500
Revenue	-
Industry branch	Energy
R&D expenses	-
Company offices	3
Brief description of product portfolio	TenneT is a leading European grid operator. We are committed to providing a secure and reliable supply of electricity 24 hours a day, 365 days a year, while helping to drive the energy transition in our pursuit of a brighter energy future – more sustainable, reliable and affordable than ever before. In our role as the first cross-border Transmission System Operator (TSO) we design, build, maintain and operate 24,500 km of high-voltage electricity grid in the Netherlands and large parts of Germany, and facilitate the European energy market through our 16 interconnectors to neighbouring countries. We are one of the largest investors in national and international onshore and offshore electricity grids, with a turnover of EUR 6.4 billion and a total asset value of EUR 32 billion.
Website	www.tennet.eu

The role of TenneT TSO GmbH is to provide a secure and reliable supply of electricity at each instant of time. In this role we are responsible to operate, built and maintain the extra-high voltage electricity network in large parts of Germany as well as in the Netherlands. The main objectives of TenneT's activities are:

- ensuring secure, reliable and economic operation of the electricity system as part of the common European power system;
- ensuring the required planning, expansion and maintenance of the transmission network and cross-border interconnections in order to facilitate the transformation towards a renewable-based system;
- making transmission capacity available to the market in order to facilitate cross-border exchanges across interconnected system;
- supporting a market development on European and national towards a renewable-based system to make efficient use of existing resources.

TenneT's general contribution to the project is to support the project partners with a practical and real-world perspective. In this role, we will contribute to designing the framework and scope of the intended solution of the project. Moreover, we expect to contribute to both preparing representative grid models as well as analyzing power flow allocation outcomes and market simulation results.

Moreover, TenneT brings in its theoretical knowledge in the field of nodal market mechanisms. During the last years TenneT investigated the potential options and challenges in designing nodal pricing system for a European context. While this approach is seen as superior in academic research, it cannot be easily adapted to the current European market framework. Identifying those challenges as well as potential design options helped to gain a better understanding of the functioning of a nodal system as well as adjust it to the characteristics of the European power system. We expect to bring in this knowledge by designing potential pricing rules as well as in the experimentation phase of the project.

2 Ziele und Arbeitsprogramm / Goals and work program

2.1 Voraussichtliche Gesamtdauer des Projekts / Expected project duration

The anticipated total duration is 2.5 years. DFG funding will be necessary for the entire duration.

2.2 Ziele und Erfolgskriterien / Goals and Success Criteria

Scientific goals of the transfer project: This project aims to develop and study different allocation and pricing rules for electricity spot markets on realistic problem sizes with realistic constraints. The results should inform policymakers and other stakeholders on how to determine future electricity market prices and design real-world electricity markets. These questions have received renewed interest due to the growing levels of renewables and the high price spikes and redispatch costs that challenge the current electricity market design. In particular, we want to address **two main questions** in this transfer project:

1. *Which optimization models best approximate the underlying AC optimal power-flow problem and allow for the computation of realistic problem sizes?* This question is of immediate relevance to the allocation in day-ahead markets.

2. *Which pricing rules best achieve design desiderata, such as low side-payments, adequate congestion signals, and low lost opportunity costs?* This question is at the core of the current market design discussions in the US and in Europe. It is important to understand differences between various nodal pricing rules and zonal prices in this context.

By addressing these questions, we aim to close a research gap and bring together two strands of literature, that of pricing with non-convexities in the economic sciences and that of optimal power flow models in engineering. We expect new insights on the interplay of clearing with transmission constraints and pricing in electricity markets and the development of new clearing and pricing rules. Apart from the immediate practical relevance of such results, we aim to get deeper insights in properties of pricing in non-convex and coupled markets and how well we can approximate design desiderata such as envy-freeness and budget balance. Apart from analytical worst-case results, the simulator will allow us to get average case results in large and realistic environments.

Planned prototype and intended outcome: The transfer project aims at a simulator based on realistic problem sizes and technical specifications, which allows stakeholders of the European electricity spot market to analyze different market designs and perform counterfactual analysis for different pricing rules for markets with large shares of renewable energies. These questions are of utmost practical importance not only for policymakers and regulatory entities, but also for transmission system operators, and they should also help reduce the rapidly growing redispatch costs that contribute to rising electricity retail prices. More importantly, adequate price signals shall incentivize demand reduction where needed, set appropriate investment incentives in low-carbon technologies, and in this way contribute to a successful decarbonization. Here, we draw on results of previous research (Ahunbay, Bichler et al. 2022, Bichler, Knörr et al. 2022).

The gap between small-scale simulations, as in our previous research, and realistic simulations is substantial. To get results with external validity, we need to consider realistic grid topologies and be able to handle the real-world constraints on the generators, the buyers, and the network. This is considerably beyond the small-scale experiments we have run so far and different from other software developed to support policymakers (see Section 2.3). To the best of our knowledge, this is **the first software simulator that allows comparisons of allocation and pricing rules, living up to TRL 6** (technology demonstrated in relevant environment). We consider the work packages defined in Section 2.3. as key milestones towards achieving the desired outcomes. We intend to make the basic software framework available as open source software such that others can build on it as well and see a demand for this.

Success criteria: A primary outcome of the project is a simulator that allows running simulations with different pricing rules, different relaxations of the ACOPF, and different data sets on realistic problem sizes. The software can be a nucleus for software offerings, and it will allow for realistic simulations that will inform policy makers and market designers about the right choices for a future-proof electricity market design. Success criteria involve the functionality of the software as described below, but also the level of new insights that we can generate about the differences among pricing systems. The latter shall be evaluated by publications in relevant electricity market journals and conferences. In addition, we are convinced that the analysis of coupled and non-convex markets will also add to the literature in economics and the management sciences.

2.3 Arbeitsprogramm inkl. vorgesehener Untersuchungsmethoden / Work program and planned investigation methods

Concept of cooperation with the application partner

The planned project is based on a close collaboration between the TUM, FIT, and the TenneT TSO GmbH. TenneT as the first cross-border TSO is an especially valuable partner for the project to reflect the requirements in different European countries. TUM provides the theoretical expertise on pricing in non-convex markets, while FIT provides in-depth electricity market know-how and software development competencies. TenneT delivers the required technical specification and they collaborate on the design and evaluation of the numerical experiments.

We plan to hire one full-time equivalent at TUM and one at FIT (TV-L E13) over the entire project period, who are closely and constantly collaborating on the individual work packages. TenneT's knowledge of the technical details of the power grid is a key enabler for the project, and it distinguishes this project from other high-level power grid simulations that are available today. Note that we need access to highly skilled technical specialists from the TSO, whose expertise is in high demand, in particular during these times. For TenneT, these are obviously very challenging times. Nevertheless, they are going to contribute to this project and commit at least nine person months at this point. This commitment was increased from the original draft proposal with two person months only and might be again increased once the turmoil on electricity markets ceases. Please see our comments at the end of this document on recent changes among the application partners. Already now, we aim for bi-weekly virtual meetings among the project team for the entire project duration.

Overall, we plan eight work packages (WP). More detailed descriptions of each work package can be found further below.

- WP 1 involves requirements engineering and a decision on the framework design of the desired simulator. The input of TenneT as key stakeholders is crucial for the definition of the scope and technical details that need to be considered. FIT leads this WP and provides expertise for a testing framework and the data layers, while TUM aggregates this input into an overarching framework design.
- WP 2 aims to compile realistic datasets. Some readily available datasets may be used for initial purposes, but TenneT's expertise is required to design representative datasets tailored to the European market and the level of technical detail that was agreed upon in WP 1. FIT leads the work in compiling datasets and preparing them for usage.
- WP 3 aims at an initial implementation of currently used clearing mechanisms, i.e., DCOPF models as well as zonal clearing algorithms. TUM takes the lead on this work package and draws on previous work. The model should reflect the level of technical details that was agreed upon in WP 1 and should be able to deal with the datasets developed in WP 2.
- WP 4 addresses the first main question regarding power flow models. Leveraging advances in non-linear convex optimization, we plan to implement the second-order conic (SOC) (Jabr 2006), the quadratic convex (QC) (Hijazi, Coffrin et al. 2017), and the semidefinite programming (SDP) (Lavari and Low 2012) relaxations. TUM will draw on open source libraries where it is useful and compile a coherent framework for the purposes laid out in WP1 and WP2.

- WP 5 shifts the focus towards pricing rules. We initially plan to implement standard Integer Programming (IP) pricing (O'Neill, Sotkiewicz et al. 2005) as it is currently used in the majority of U.S. markets as well as Extended Locational Marginal Pricing (ELMP) (MISO 2019) as an established approximation of Convex Hull (CH) pricing (Hogan and Ring 2003, Gribik, Hogan et al. 2007). TUM will lead this WP.
- WP 6 aims at implementing a novel pricing rule from TUM's previous research, the so-called Join (Ahunbay, Bichler et al. 2022). Unlike previous pricing rules, the main idea of the Join is to jointly consider minimal make-whole payments and correct congestion signals in the price determination. TUM will focus on the implementation.
- WP 7 comprises the development of an experimental design and of regression tests to ensure that the developed simulator meets the desired standards. With this work package, we want to ensure that the outcome of the project has external validity. FIT will coordinate this WP and draw on the TenneT's input and expertise as stakeholders.
- WP 8 is devoted to an extensive experimental evaluation of the implemented clearing and pricing rules. We hope to gain new insights on the scalability of algorithms and properties of prices on realistic and representative datasets. FIT and TenneT will jointly lead this WP.

Work package (WP)	Year 1	Year 2	Year 3	TUM	FIT	TenneT
WP 1: Framework design	**			2 PM	2 PM	2 PM
WP 2: Preparation of realistic problem instances	****			3 PM	5 PM	1 PM
WP 3: Development of linear approximations (DCOPF, zonal)	****	**		4 PM	2 PM	0 PM
WP 4: Development of convex relaxations (SOC, QC, SDP)		****		4 PM	3 PM	0 PM
WP 5: Development of IP pricing and ELMP pricing rules		****		7 PM	3 PM	1 PM
WP 6: Development of the Join price rule		**	**	6 PM	4 PM	1 PM
WP 7: Development of a framework for regression testing and numerical experiments	**	****		2 PM	5 PM	1 PM
WP 8: Experimental evaluation of different pricing rules			**	2 PM	6 PM	3 PM

Stars indicate the quarters of a year

Work Package 1: Framework Design

The intended outcome of this transfer project is a simulator that can be a nucleus for commercial offerings and used for academic purposes. In this work package, we aim to determine and specify the functionalities and architecture that are to be implemented in the subsequent work packages. While the core functionality of our simulator is the simultaneous consideration of various clearing and pricing rules, further details need to be worked out and aligned between the project partners.

In particular, we want to discuss the software architecture, i.e., how to implement packages and layers to enable functionality, maintainability, and testability of the software. We aim to adhere to established software design principles (Martin 2008). The architecture is also based on detailed requirements engineering and an agreement on the level of technical detail that we pursue. In this context, the input of TenneT is essential. For instance, we want to decide on the

modeling scope of market participants (e.g., which technical generator constraints need to be considered) and the transmission network (e.g., which network control elements need to be considered). We aim to align the input of TenneT with theoretical considerations provided by TUM and FIT and to allow for a realistic representation of European electricity systems. The output of this work package is a detailed model of the software that is aligned between all project partners and to be implemented in the following work packages.

Work Package 2: Preparation of realistic problem instances

Based on the outcomes of WP 1, we aim to compile realistic data sets for testing purposes during the development of the software and for experimental evaluations once the software has been implemented. In particular, data is needed on supply- and demand-side bids and on network specifications, ideally over a long time period. Some simplified data sets (e.g., IEEE test networks or ARPA-E Grid Optimization Challenge⁵) can serve as a good starting point. However, these data and technical specifications are representative of U.S. electricity markets. The main focus of our project, however, is to derive insights on European electricity markets. We therefore draw on the expertise of TenneT to design representative data sets that align with the requirements set in WP 1. Therefore, this work package involves coordination between the project partners, data parsing, cleaning, preparation, and validation.

Work Package 3: Development of linear approximations

WP 3 focuses on the implementation of first software components. As an outcome of this WP, we aim to replicate common clearing mechanisms as they are currently used in electricity markets, in line with the scope and architecture defined in WP 1. In particular, we want to implement both nodal and zonal market clearing mechanisms. European electricity markets are currently cleared as zonal markets, but there are ongoing discussions on future market designs that involve nodal clearing at least within certain geographical regions (Ashour Novirdoust, Bhuiyan et al. 2021). In order to contribute to the discussion on zonal or nodal pricing, we want to consider both market designs in this project.

For the nodal market design, we plan to implement a standard DCOPF problem with linearized transmission constraints (Molzahn and Hiskens 2019). The DCOPF makes simplifying assumptions for power flow (e.g., no line resistances and reactive power), but its linearization makes it highly scalable, which is why it is used world-wide. TUM has used DCOPF implementations in previous research (Ahunbay, Bichler et al. 2022, Bichler, Knörr et al. 2022) and can draw on respective mixed-integer linear programming implementations. Of course, the implementations may need to be aligned and extended to meet the requirements set in WP 1. For the zonal market design, we aim to replicate the Euphemia algorithm (Committee 2020) that is currently used for market clearing and pricing in European electricity markets. We may adapt some of the functionalities in anticipation of future market design revisions (e.g., removal of PUN orders). We accompany the implementation by initial tests based on the data compiled in WP 2.

Work Package 4: Development of convex relaxations

WP 4 goes beyond standard implementations and addresses the first research question. While there are ambiguous empirical findings on the accuracy of DCOPF (Molzahn and Hiskens 2019), recent findings suggest that tighter convex relaxations of ACOPF could bring significant upsides (Cain and O'Neill 2012, Larrahondo, Moreno et al. 2021).

⁵ <https://gocompetition.energy.gov/challenges>

We want to leverage recent advances in convex non-linear optimization and implement various alternative power flow relaxations. This includes the second-order conic (SOC) (Jabr 2006), the quadratic convex (QC) (Hijazi, Coffrin et al. 2017), and the semidefinite programming (SDP) (Lavaei and Low 2012) relaxations. This enables us to benchmark different power flow models with respect to their underlying market clearing. We are especially interested in the trade-off between reduced redispatch costs and computational cost that is to be expected by tighter power flow models. We will draw on available open source libraries for optimization and electricity market modeling where this is useful, but aim to embed the different components into a single framework and calibrate the models as planned in WP 1 and 2.

Work Package 5: Development of IP pricing and ELMP pricing rules

Besides market clearing and different power flow models, we want to examine different pricing rules and their properties on large markets of realistic problem sizes. In WP 5, we implement existing pricing rules. While electricity pricing has a long history with many pricing rules proposed over the years (Liberopoulos and Andrianesis 2016), two pricing rules are particularly relevant in current nodal U.S. electricity markets. Integer Programming (IP) Pricing (O'Neill, Sotkiewicz et al. 2005), often referred to as standard Locational Marginal Pricing (LMP), is the basis for most market designs applied today. It allows for accurate congestion signals (assuming linearized power flow models), meaning a price difference between two nodes only occurs if the network exhibits congestion. However, uplift payments can vary substantially and might be very high. Another natural choice for a pricing rule is CH Pricing (Hogan and Ring 2003, Gribik, Hogan et al. 2007). Its main idea is to replace each non-convex valuation or cost function by its convex envelope. The resulting prices minimize the incentives for market participants to deviate from the welfare-maximizing allocation, measured by so-called lost opportunity costs. If the market was convex, a Walrasian equilibrium would exist, and lost opportunity costs would be zero. However, since CH pricing is computationally hard (Schirotter, Zheng et al. 2016), so-called ELMP (MISO 2019) is used as an approximation. Under certain assumptions, ELMP is equivalent to CH Pricing (Hua and Baldick 2017), but this is generally not the case for more complicated bidding formats.

In this WP, we will implement these pricing rules. We want to analyze the resulting prices in some initial experiments and are especially interested in the behavior of the pricing rules under different convex power flow models.

Work Package 6: Development of the Join pricing rule

In WP 6, we plan to go beyond the standard pricing rules defined in WP 5 and address the second research question. TUM has recently conducted research on novel pricing rules that balance trade-offs in non-convex electricity markets in a more principled manner. In particular, while CH Pricing minimizes lost opportunity costs, the actual side-payments that are paid out to market participants are actually much lower. The side-payments, also referred to as make-whole payments, typically only cover for losses that market participants may incur. In contrast, lost opportunity costs also capture higher forgone profits of participants. Since growing levels of make-whole payments are a concern on electricity markets, TUM suggested a pricing rule that explicitly addresses these concerns (Bichler, Knörr et al. 2022).

Our dual pricing framework for non-convex electricity markets allows for a principled way to balance trade-offs in the design of payment rules. In particular, it has been observed that – besides make-whole payments – the quality of the congestion signals is of essential importance, since

this affects the congestion income distribution and the remuneration of hedging contracts and other financial products. The so-called Join Pricing rule (Ahunbay, Bichler et al. 2022) jointly optimizes make-whole payments and congestion signals and has shown promising results on small test networks based on DCOPF. In WP 6, we want to incorporate these pricing rules into our simulator. Previous implementations require adaptations to fit the different convex power flow relaxations from WP 4 and the requirements set forth in WP 1. WP 6 could also include the implementation of other alternative pricing rules arising from the project partners' research or recent publications.

Work Package 7: Development of a framework for regression testing and numerical experiments

In parallel to the implementations discussed for the previous WPs, we want to explicitly designate a WP to quality assurance for our implementations and to prepare for extensive numerical experiments. The latter are necessary to ensure that the implemented software meets its requirements at the end of the project. In particular, we want to develop a framework for regression testing to ensure the integrity and quality of our programs and the changes we make. We actively draw on the expertise of TenneT to validate our results and want to specify meaningful test cases that have external validity. We consider this to be an important WP to ensure that the final product allows for reliable comparisons of clearing and pricing rules. At the same time, in this WP we want to draft an experimental design as a preparation for WP 8 and to provide feedback for WPs 3-5.

Work Package 8: Experimental evaluation of different pricing rules

WP 8 brings together all previous WPs and it shall generate results about pricing rules that are relevant for academia, policy makers, and other stakeholders like TenneT. Via extensive numerical experiments, we aim to provide empirical support for our two main research questions and further test the capabilities of our simulator. We envision a modular structure where different data sets can be combined with different power flow representations and different pricing rules. To our knowledge, this is the first simulator to unite all of these aspects. We hope that with our findings, we can gain insights on the scalability of different algorithms and the properties of allocation and prices for real-world power networks. We anticipate that the convex non-linear programs will require substantial computation power, and we plan to use computing resources at the Leibniz Supercomputing Center in Garching.

2.4 Verwertung / Exploitation

Scientific goals and exploitation: As indicated earlier, pricing in electricity markets has been a topic of debate for more than 20 years (Liberopoulos and Andrianesis 2016). While the promise of market mechanisms to allocate scarce resources rests on fundamental results such as the welfare theorems (Arrow and Debreu 1954), it has been shown that Walrasian equilibria only exist under very restrictive assumptions. We worked on new proposals for electricity pricing that address the resulting problems (Ahunbay, Bichler et al. 2022, Bichler, Knörr et al. 2022) and aim to use this project to develop these pricing rules further, to validate them on realistic networks representative of the European market, and to generate policy advice and new research ideas based on our findings that arise over the course of the project. We discussed the main scientific questions and expected contributions in Section 2.2. We are convinced that solutions to these problems are well suited for highly ranked journals in the management sciences such as Operations Research, Information Systems Research, Energy Economics, and the European

Journal of Operational Research. We also plan to attend international conferences such as the INFORMS or the EURO Annual Meeting as well as more specific meetings on electricity market design such as EEM.

Commercial relevance of the problem and impact: The European day-ahead electricity market is the largest liberalized electricity market worldwide. European market coupling has led to a single centralized clearing mechanism that covers 98.6% of EU electricity consumption, amounting to around 1.530 TWh per year. In 2021 this led to €200 million average daily value of matched trades. The allocation and pricing mechanisms are under scrutiny today, due to strongly increasing redispatch costs. Rising redispatch costs and significant curtailment of renewables underline the economic, ecologic, and political pressure to find new market designs in Europe and Germany. Electricity spot market design has made recent headlines due to statements of leading politicians in Germany and in the EU to find adequate designs for future electricity markets with 100% renewables. Our results shall provide guidance to policy makers in this respect, which may also help to further reduce both carbon emissions and the dependence on foreign resource imports like Russian gas.

Target market for commercial exploitation: Even without the current political debate about high wholesale prices, the European spot market design would be likely to see significant changes due to the limits of the current zonal pricing model. Current European debates for instance take place within the BZR or the consultation of the reform of the EU electricity market design. Here, the European Commission (EC) sees regional price incentives as part of the solution for reducing electricity costs⁶ and the European Network of Transmission System Operators for Electricity (ENTSO-E) explicitly names nodal pricing in its marked design vision⁷. Market innovations were also demanded in Germany's Annual Economic Report of the Federal Ministry for Economic Affairs and Climate Action and are heavily discussed within the Platform Climate Neutral Electricity System⁸. Even countries with a high proportion of "controllable" energy resources, such as the UK⁹, are discussing the introduction of nodal pricing, which marks a significant change in the electricity market design.

The CACM 2.0 amendments on the Capacity Calculation and Congestion Management Regulation proposes non-uniform prices as a next step. Over the next few years, policymakers and TSOs are asked to provide input on the question how non-uniform prices can be implemented. Today, linearized power flow models are under scrutiny due to associated welfare losses.

Our project can provide direct input for the corresponding policy questions. Policymakers (e.g., the EC and ministries), regulators (e.g., the Bundesnetzagentur), power exchanges (e.g., EPEX Spot or NordPool), or TSOs need to study and understand the impact of such market design changes. However, these questions are also of central importance to 42 TSOs in the EU. Experiments to run counterfactual analyses are a central approach to addressing these questions.

Competition in the target market: A number of institutions (e.g., consentec, FfE Munich, the Institute of Energy Economics at the University of Cologne (EWI), Öko-Institut e.V., or prognos) have developed high-level market simulators in the past. To our knowledge, these products are based on linear or mixed-integer programming and simplified power grid models. As far as we know, none of these solutions uses state-of-the-art convex relaxations and none of the tools allows for the analysis of different pricing rules. The main purpose of existing solutions is to run

⁶ <https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:52022DC0236&from=EN>

⁷ https://eepublicdownloads.entsoe.eu/clean-documents/Publications/Market%20Committee%20publications/210331_Market_design%202030.pdf

⁸ <https://www.bmwk.de/Redaktion/EN/Publikationen/Wirtschaft/annual-economic-report-2022.html>

⁹ <https://bidstats.uk/tenders/2021/W48/764084401>

analyses regarding different demand and supply scenarios (e.g., renewable energy supply). The purpose of our software is different, as we want to be able to identify differences of specific market design rules and recommend adequate allocation and pricing rules to be implemented. Especially the quest for new non-uniform pricing rules in CACM 2.0 mentioned earlier will lead to significant demand for consulting services by policymakers, regulators, and TSOs.

Future offerings for commercial exploitation: A software based on the results of this project shall be used in publicly funded projects and in consulting projects for the organizations mentioned above. As described before, FIT BISE plays a key role in the Kopernikus-Project SynErgie and regularly conducts consulting projects to provide advice for TSOs and other industry partners (FIT BISE is characterized by a high share of industry-funded projects). Such projects describe engagements for a group of Fraunhofer researchers. The existing industry know-how combined with the electricity market simulator is a unique selling proposition which enables to enter a market that cannot be served now. Given the assumed consulting demand described above with TSOs, power exchanges, regulators, and policymakers, in addition to the publicly funded projects and our policy advice we expect to acquire about 10 to 20 consulting projects between the third and the eighth year after the project start, each with average revenues of around 300,000 € (two Fraunhofer researchers over around six months as consultants). In addition, the software could also be licensed in the future, which can lead to further revenues. Licensees of the software could use the simulator for individual analyses and as a decision-support tool. The software could also serve as a basis to develop customized solutions tailored to the clients' requirements and needs.

How the project helps in developing future offerings: The transfer shall lead to a simulation engine that scales to realistic problem sizes and can, in principle, handle the types of constraints and technical specifications that one would encounter in practice in Germany and Europe. We plan to focus on the backend, not on visualizations and reporting functions required for a consulting offer. A future final product could also support, for example, (1) the design of network charges with local components, (2) be used as a basis for the analysis of inc-dec gaming, (3) answer questions related to market granularity/timing throughout Europe, (4) analyze interactions between zonal and nodal markets, (4) make comparisons with forms of average pricing, or (5) evaluate state interventions like price caps. The project will help FIT to design a nodal pricing simulator that scales to national power networks and derives outcomes with high external validity. TenneT will be crucial to ensure that the simulator lives up to its expectations and that the software can handle real-world requirements. TenneT participates in the project to access the software early on.

2.5 Umgang mit Forschungsdaten / Handling research data

We plan to test and benchmark our clearing and pricing rules on several realistic data sets. As discussed in the context of WP 2, we will draw on publicly available datasets (e.g., IEEE test networks or data from the ARPA-E Grid Optimization Challenge). These datasets are well-documented and suitable for replication. In addition, we plan to create realistic data sets from publicly available data (e.g., provided by den ENTSO-E in the context of the bid zone review). We also plan to align the access to our used data with German and European data spaces initiatives like Gaia-X or Energy-X and coordinate with relevant NFDI projects.

Regarding our software, Fraunhofer will release the software framework by means of a GitHub repository as open-source software under the BSD license with a detailed documentation once the packages are tested. This way, we want to ensure that our findings and numerical results in

our scientific publications are reproducible. Additional components for visualization and data analysis that will be required for later consulting engagements and the commercialization strategy (see Section 2.4.) are not part of this project where we focus on the backend simulation engine.

2.6 Relevanz von Geschlecht und/oder Vielfältigkeit / Relevance of gender and/or diversity

We are committed to diversity and will at any time aim for gender balance and the DFG's diversity guidelines. Both TUM and FIT have programs in place and aim for a project team that is balanced and represents different genders, origins, age, etc. Further information on the team composition can be found in Section 4.4. Notwithstanding the relevance of diversity, this section is on the question whether gender and/or diversity are relevant for the subject of this research project (not the team composition). Given the nature of the research questions that focus on allocation and pricing on electricity market, we argue that this is not the case.

3 Projekt- und themenbezogenes Literaturverzeichnis | Patente / Bibliography | Patents

3.1 Projekt- und themenbezogenes Literaturverzeichnis / Bibliography

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3.2 Patente / Patents

3.2.1. Angemeldet / Registered

None. Algorithms cannot be patented in Europe.

3.2.2. Erteilt / Granted

None. Algorithms cannot be patented in Europe.

Patentrecherche / Patent research

Since algorithms cannot be patented in Europe, our research and simulator will not be affected by any other patents. We searched the database of the European Patent Office (Espacenet) with various search strings. We found some patents issued in China that refer to electricity pricing, yet none of them are relevant for our commercialization strategy in Germany and Europe. The search strings included, for instance, "electricity pricing", "non-convex pricing", "pricing algorithm", "optimal power flow", or "power flow relaxation".

4 Begleitinformationen zum Forschungskontext / Accompanying information on the research context

4.1 Angaben zu ethischen und/oder rechtlichen Aspekten des Vorhabens / Information on ethical and/or legal of the project

4.1.1. Allgemeine ethische Aspekte / General ethical aspects

The scope of our project involves theoretical work and software implementation and, therefore, does not pose any risks and/or burdens for certain persons or group of people. We do not expect any negative effects. The outcome of our work should enable policy advice for decisions on future European electricity market design, but our work will not have immediate political consequences. We will continuously monitor if risks arise and react accordingly.

4.1.2. Erläuterungen zu den vorgesehenen Untersuchungen bei Versuchen an Menschen, an vom Menschen entnommenem Material oder mit identifizierbaren Daten / Explanations of the investigations envisaged in the case of tests on humans, on material taken from humans or with identifiable data

No such experiments are conducted and no identifiable human data will be used.

4.1.3. Erläuterungen zu den vorgesehenen Untersuchungen bei Versuchen an Tieren / Explanations on the planned for experiments on animals

No such experiments are conducted.

4.1.4. Erläuterungen zu Forschungsvorhaben an genetischen Ressourcen (oder darauf bezogenem traditionellem Wissen) aus dem Ausland / Explanations on research projects on genetic resources (or traditional knowledge related to them) from abroad

No such projects are conducted.

4.1.5. Erläuterungen zu möglichen sicherheitsrelevanten Aspekten („Dual-Use Research of Concern“, Außenwirtschaftsrecht) / Explanation of possible security-relevant aspects ("Dual-Use Research of Concern", foreign trade law)

There are no security-relevant aspects to our research.

4.2 Angaben zur Dienststellung / Position information

University

Prof. Dr. Martin Bichler, Full Professor (W3), School of Computation, Information and Technology, Technical University of Munich

Fraunhofer

Prof. Dr. Martin Weibelzahl - Group Leader "Sustainable Energy Management & Mobility" and Part of the Management Team at the Branch Business & Information Systems Engineering of the Fraunhofer Institute for Applied Information Technology FIT (FIT BISE)

4.3 Angaben zur Erstantragstellung / Information on the initial application

Not applicable.

4.4 Zusammensetzung der Projektarbeitsgruppe / Team composition

Next to the applicants (Prof. Bichler – University; Prof. Weibelzahl – Fraunhofer), the project team consists of the two funded doctoral researchers (TV-L E13) as well as student assistants, and employees of TenneT. The key contact person at TenneT is Dr. Christoph Neumann.

Regarding the cooperation within the project, gender and diversity aspects are taken seriously at FIT/FIT BISE and at TUM. We will make sure that the project team is balanced and represents different genders, origins, age, etc. as mentioned in Section 2.6. For instance, the impact of continuous measures of FIT BISE to increase diversity is particularly evident in the increase in the proportion of women among its researchers in recent years, having increased from 17.6% in 2018 to 32.63 % in 2023. In pursuit of the goal to promote women, a comprehensive career and development program (consisting, among other things, of the appointment of equal rights officers and the planning of regular events on the topic of Family & Career) has been developed. FIT BISE also has several female post-docs with a habilitation goal and female professors in its management team to foster diversity.

TUM affirms continued commitment to the principle of an open-minded university that attracts and supports the best talents in all their diversity. Equality is an elementary pillar of TUM, and the university is committed to a focus on diversity and talent management. In December 2021, the TUM published a Gender Equality Plan¹⁰ which represents TUM's continued commitment to the principle of an open-minded university that attracts and supports the best talents in all their diversity. These measures are intended to create the best possible conditions for each person to develop their individual potential. TUM has set itself the target to become Germany's most attractive university for female students and scientists. Equal opportunity is anchored in TUM's General Statute (Grundordnung), in its Mission Statement, the university's development plan as well as in its target agreements with the Ministries of Science and Education. Furthermore, the topic is included in the TUM Faculty Tenure Track, the TUM Graduate School Statute and the Mission Statement. We will build on and further develop the described measures to foster diversity in our project and ensure that each step of the collaboration (including the hiring step) will be free of discrimination.

4.5 Zusammenarbeit mit Wissenschaftlerinnen und Wissenschaftlern in Deutschland in diesem Projekt / Cooperation with scientists in Germany in this project

A cooperation with researchers from other research institutions beyond the ones mentioned in this proposal is not intended. As a result of the involvement in the Kopernikus SynErgie project, there are good and established contacts with various other German research institutes (e.g., DIW, or ZEW), which may be leveraged if this is required. However, the project goals in this transfer project are distinct from those in SynErgie.

¹⁰ <https://mediatum.ub.tum.de/doc/1639267/1639267.pdf>

4.6 Zusammenarbeit mit Wissenschaftlerinnen und Wissenschaftlern im Ausland in diesem Projekt / Cooperation with scientists abroad in this project

As a result of the involvement in the Kopernikus SynErgie project and through personal contacts, there are good and established contacts with various other research institutes (e.g., IRENA or NHH) and researchers from the United States (e.g., Richard O'Neill, Benjamin Hobbs). These contacts may be leveraged if this is deemed useful.

4.7 Wissenschaftlerinnen und Wissenschaftler, mit denen in den letzten drei Jahren wissenschaftlich zusammengearbeitet wurde / Scientists with whom scientific collaboration has taken place in the last three years

Martin Bichler has collaborated with Hans Ulrich Buhl (FIM Research Center for Information Management and Branch Business & Information Systems Engineering of the Fraunhofer FIT), Peter Gritzmann (TUM), Stefan Minner (TUM), Veronika Grimm (FAU Erlangen-Nürnberg), Paul Klemperer (Oxford University), Paul Milgrom (Stanford University), Thayer Morrill (NCSU), and Stefan Waldherr (VU Amsterdam).

Martin Weibelzahl has collaborated with Mette and Endre Bjørndal (NHH), Hans Ulrich Buhl (FIM Research Center for Information Management and FIT BISE), Stefano Coniglio (University of Bergamo), Gilbert Fridgen (University of Luxembourg), Veronika Grimm (FAU Erlangen-Nürnberg), Raphael Heffron (Universite de Pau et des Pays de l'Adour), Stefan Waldherr (VU Amsterdam), and Gregor Zöttl (FAU Erlangen-Nürnberg).

4.8 Projektrelevante Zusammenarbeit mit erwerbswirtschaftlichen Unternehmen / Project-relevant cooperation with commercial enterprises

The cooperation agreement according to the Fraunhofer and DFG model cooperation agreement is attached to this project proposal.

4.9 Projektrelevante Beteiligungen an erwerbswirtschaftlichen Unternehmen / Project-relevant participations in commercial enterprises

There are no project-relevant investments in commercial enterprises.

4.10 Apparative Ausstattung / Equipment

All project partners have a number of laptops and computing servers which can be used by the staff for research projects, in particular also for the project under consideration. For the expected high-computing effort for the non-linear power flow relaxations, we may use computing resources at the Leibniz Supercomputing Center in Garching.

4.11 Weitere Antragstellungen / Further applications

An application for funding of this project has not been submitted to any other agency. If the applicant submits such an application, the DFG will be notified immediately.

4.12 Weitere Angaben / Further information

We commit to comply with the rules of good scientific practice when submitting the application for this grant to the DFG.

5 Beantragte Module/Mittel / Requested modules/funds

Finanzplan / Financial Plan

	DFG	Fraunhofer (ohne AfA)	Anwendungspartner TenneT
Personnel funds	1 TV-L E13 position 30 months € 188,178	[inkl. Personalgemein- kosten] 1 TV-L E13 position 30 months € 357,364	€ 0
Equipment up to 10.000 Euros (DFG funding only), consumables	€ 0	€ 0	€ 0
Travel expenses	€ 4,100	€ 4,100	€ 0
Other expenses (Fraunhofer: e.g. third-party services)	€ 0	€ 0	€ 0
Publication funds	€ 1,000	€ 0	€ 0
Investment funds (only DFG applicant)	€ 0	Keine Beantragung von Investitionsmitteln zulässig	€ 0
Further modules if applicable	€ 3,000	€ 0	€ 0
Total	€ 196,278	€ 361,464	€ 0¹¹

Detaillierte Darstellung der Finanzierung durch die DFG / Detailed description of the funding by the DFG

5.1 Basismodul / Basis module

5.1.1. Personalmittel / Personnel funds

To conduct the proposed research program, TUM and FIT request funding to each hire one doctoral researcher for the project duration of 30 months. They will be paid according to the TV-L E13 pay scale. Given the nature and complexity of the research, we require students with a computer science, information systems, mathematics, or electrical engineering background. In

¹¹ see our comments at the end of this document on details of the contribution by TenneT.

particular, we require solid software development skills to develop our desired simulator. For these reasons, we ask for fully funded positions.

5.1.2. Sachmittel / Material resources

5.1.2.1. Geräte bis 10.000 Euro, Software und Verbrauchsmaterial / Equipment up to 10,000 Euros, software and consumables

TUM and FIT can provide laptops, servers, and other equipment for the hired doctoral researchers.

5.1.2.2. Reisemittel / Travel funds

As part of their research and PhD program, the doctoral researchers at TUM and FIT should attend two to three conferences in the second and third year of the project and present their research to the scientific community. Covering traveling expenses, accommodation and conference fees, we estimate ~1,250 EUR for an overseas conference and ~1,250 EUR for the conferences in Europe. Therefore, we ask for a travel budget of 2,500 EUR in total for each doctoral researcher.

5.1.2.3. Mittel für wissenschaftliche Gäste (ausgenommen Mercator-Fellow) / Funds for scientific guests (except Mercator Fellows)

We plan to have at least two in-person meetings of all project partners in each project year (in Augsburg/Munich/Bayreuth, six meetings in total). For the meetings that involve traveling, we estimate train costs of 100 EUR and accommodation costs of 300 EUR for TUM and the same amount for FIT (each for one doctoral researcher and one PI). Over the project period, this sums up to 1,600 EUR for TUM and FIT each.

5.1.2.4. Mittel für Versuchstiere / Funds for laboratory animals

No such funds apply.

5.1.2.5. Sonstige Mittel / Other funds

No such funds apply.

5.1.2.6. Publikationsmittel / Publication funds

For the entire project duration, we request 1,000 EUR in publication funds. This covers proof-reading services and submission fees to high-ranking journals.

5.1.3. Investitionsmittel / Investment funds

5.1.3.1. Geräte über 10.000 Euro / Equipment over 10,000 euros

No such funds apply.

5.1.3.2. Großgeräte über 50.000 Euro / Large equipment over 50,000 euros

No such funds apply.

5.2 Modul Eigene Stelle / Module own position

No such funds apply.

5.3 Modul Vertretung / Module substitution

No such funds apply.

5.4 Modul Rotationsstellen / Module rotation positions

No such funds apply.

5.5 Modul Mercator Fellow / Module Mercator Fellow

No such funds apply.

5.6 Modul Projektspezifische Workshops / Module project-specific workshops

No such funds apply.

5.7 Modul Öffentlichkeitsarbeit / Module public relations

In the second and third year of the project, we request 3,000 EUR for a web presence and public relations. With these funds, we intend to enhance the transfer of our project results to practice.

5.8 Modul Pauschale für Chancengleichheitsmaßnahmen / Module for equal opportunity measures

No such funds apply.

Finanzierung durch Fraunhofer / Funding by Fraunhofer

Scientific employees with the qualification of a master's degree are required to carry out the project contents. Because of their comprehensive expertise and methodological competence, they are crucial for the implementation of the various work packages in line with current scientific standards and, overall, for the successful completion of the project. More precisely, the project includes, amongst others, the development of electricity market models that can handle realistic problem sizes, the formulation and analysis of different pricing rules, as well as an evaluation of the derived pricing rules in numerical test cases. Therefore, the employment of scientific staff is required, who will be paid according to TV-L E13 and according to the official cost calculation by Fraunhofer.

Beitrag des Anwendungspartners / Contribution of the application partners

The contribution of TenneT TSO GmbH is in providing an applied perspective to the project. TenneT provides us with technical knowledge about the power grid, which will be crucial for the model formulations and the preparation of new data sets. Importantly, they will provide feedback on the experiments to make sure that the results of the simulations have external validity. These tasks require colleagues with deep technical expertise about the power grid and the power management and dispatch procedures currently in use. Personnel contributions as well as other costs (e.g., travelling and equipment) are carried by TenneT TSO GmbH for its own activities. The 9 PMs have an estimated value of around EUR 140,000 including overhead as well as travel expenses.